Chapter 2 — Creating Date & Time Objects

Before working with dates you need to create them. This chapter covers every method for creating individual timestamps, date ranges, period ranges, and timedelta ranges.

2.1 pd.Timestamp()

pd.Timestamp is the single-value date/time object in pandas. You can construct it from strings, keyword arguments, or existing Python datetime objects.

Creating Timestamps

```
# From a string
ts1 = pd.Timestamp('2024-07-15 14:30:45')
print(ts1)

# From keyword arguments
ts2 = pd.Timestamp(year=2024, month=7, day=15, hour=14, minute=30, second=45)
print(ts2)

# From a Python datetime
from datetime import datetime
ts3 = pd.Timestamp(datetime(2024, 7, 15, 14, 30, 45))
print(ts3)

# Current time
ts4 = pd.Timestamp.now()
print(ts4)
```

Output:

```
2024-07-15 14:30:45
2024-07-15 14:30:45
2024-07-15 14:30:45
2024-07-15 14:30:45.123456   # (actual current time)
```

2.2 pd.to_datetime()

This is the most-used conversion function. It handles single values, lists, Series, and entire DataFrame columns.

Converting Various Inputs

```
# Single string
print(pd.to_datetime('2024-07-15'))

# List of strings
dates = pd.to_datetime(['2024-01-01', '2024-06-15', '2024-12-31'])
print(dates)

# DataFrame column
df = pd.DataFrame({'date_str': ['2024-01-01', '2024-06-15', '2024-12-31'],
                   'value': [100, 200, 300]})
df['date'] = pd.to_datetime(df['date_str'])
print(df.dtypes)

# With custom format
pd.to_datetime('15-07-2024', format='%d-%m-%Y')
```